

Do Explainers Agree on Which Features Matter? Instance-Level Agreement between SHAP and LIME on Three Tabular Datasets

Jonathan Herrera-Vásquez*

Universidad Americana de Europa (UNADE), Department of Computer Science, Cancún,
Mexico

jonnabio@gmail.com

Abstract. Post-hoc explainers are often used as if they were interchangeable, yet two explainers may name different features for the same prediction. This study measures how much SHAP and LIME agree at the level of the single instance. Both methods explained the same instances of the same models on three tabular datasets (Adult, German Credit and Breast Cancer Wisconsin), giving 31,411 paired explanations in 86 runs. The analysis plan was registered before any agreement statistic was computed; runs, not instances, are the unit of summary, and intervals resample random seeds. The mean overlap of the five most important features (Jaccard index) was 0.393 on Adult, 0.387 on German Credit and 0.714 on Breast Cancer, well above the overlap of random feature sets but far from identity, and the ordering of the features agreed weakly on two of the three datasets. Agreement depended on the model family and did not depend on the number of explained instances. It was not consistently lower for misclassified instances. Disagreement was larger where the fidelity of the LIME explanation was lower. The agreement of signs was low and followed the predicted class. A post hoc analysis shows why: LIME without discretisation returns local slopes, while SHAP returns signed contributions. After the slopes were converted into contributions, the two methods agreed on the sign of 0.945 to 0.990 of the shared top features. The two explainers therefore differ in which features they rank first, not in the direction they assign to them.

Keywords. Explainable artificial intelligence, post-hoc explanations, disagreement problem, feature attribution, SHAP, LIME.

1 Introduction

Post-hoc explanation methods describe a prediction of a trained model in terms of its input features. SHAP [17] and LIME [20] are the most used for tabular data: each returns, for one instance, a signed weight per feature. In practice the two are treated as alternatives, and the choice between them is made on cost or habit. That practice assumes that the explanation a user sees does not depend much on the method.

The assumption has been questioned. Krishna et al. [15] named the *disagreement problem*: explanation methods often identify different top features for the same prediction, and practitioners have no principled way to resolve the conflict. Similar observations were made for text models [19] and for defect prediction [22]. If the features shown to a user change with the explainer, an explanation is a property of the pair (model, explainer), not of the model alone, and conclusions drawn from it need the same caution [7, 23].

Most evidence on disagreement comes from a small number of models per dataset and treats the explained instances as independent observations. Two points remain open for tabular classifiers. The first is how much agreement varies with the model family, the dataset and the number of explained instances when the same instances are explained by both methods. The second is whether disagreement is related to two properties that practitioners care about: whether the prediction is

correct, and how good each explanation is by the usual quality measures.

This study addresses both points with instance-paired explanations from an existing benchmark cohort and a new, instance-aligned cohort produced for this purpose. The research questions are:

- **RQ1.** For the same instance and model, how much do SHAP and LIME agree on the most influential features and on their order?
- **RQ2.** How does this agreement vary with the model family, the sampling intensity and the dataset?
- **RQ3.** Is agreement different for correctly classified and misclassified instances?
- **RQ4.** Is disagreement on an instance associated with the fidelity or the stability of either explanation?

A secondary, descriptive question is how the features used by two explainers of a different kind, Anchors [21] and DiCE [18], overlap with those of SHAP and LIME.

The contributions are: (i) estimates of SHAP–LIME agreement on 31,411 paired explanations across five model families and three datasets, with the run as the unit of summary and intervals clustered by random seed; (ii) evidence that agreement depends on the model family and the dataset, not on the number of explained instances, and is not consistently related to prediction correctness; (iii) an association between disagreement and the fidelity of the LIME explanation; and (iv) a methodological finding: the sign agreement between SHAP and LIME without discretisation is determined mostly by the predicted class, because the two methods return different quantities, and it becomes high once both are expressed as contributions.

2 Related Work

Disagreement between explainers. Krishna et al. [15] formalised disagreement with measures of feature agreement, rank agreement, sign agreement and rank correlation over the top- k features, and reported frequent disagreement across datasets and models, together with interviews showing that practitioners resolve it with ad hoc rules. Neely et al. [19] found low rank correlation between attribution methods for text classifiers and argued that agreement should not be used to validate a method. Roy et al. [22] reported disagreement between LIME and SHAP for defect prediction models, on the features and on their sign.

Why methods differ. Han et al. [10] showed that several attribution methods are local function approximations of the model that differ in the neighbourhood and the loss they use, so that no method is best for every neighbourhood. Garreau and von Luxburg [9] analysed LIME for tabular data and showed that, for a linear model, its coefficients are proportional to the gradient of the model. SHAP values, in contrast, distribute the difference between the prediction and an expected prediction among the features [16, 17]. This difference of meaning matters for the sign analysis in Section 4.3.

Reliability of explanations. Explanations can change under small perturbations of the input [1] and can be manipulated [24]. Stability and fidelity are therefore common quality criteria. This study relates them to disagreement at the level of the instance, which, to our knowledge, earlier work on tabular data has not reported with a clustered design.

3 Materials and Methods

3.1 Datasets, Models and Runs

Three public binary classification datasets were used: Adult [2] (108 encoded features), German Credit [12] (61 encoded features) and Breast Cancer Wisconsin, diagnostic [25] (30 numeric features). Numeric features were standardised and categorical features were one-hot encoded; each encoded column is one feature in every analysis.

On Adult, five model families were explained: logistic regression (LR), a support vector machine with probability outputs (SVM), a multilayer perceptron (MLP), a random forest (RF) [4] and gradient boosting (XGB) [6]. Each explainer was run with five random seeds and three sampling intensities ($n = 50, 100$ and 200 test instances per quadrant of the confusion matrix), so that correct and incorrect predictions are both represented. Each Adult model was trained once, on the partition of the data given by seed 42; the seed of a run fixes the partition from which its instances are drawn. For the four seeds other than 42 that partition differs from the one used in training, so part of the explained instances belong to the training data of the model. On German Credit and Breast Cancer, RF and XGB models were trained with three seeds each, on the partition of the same seed, and explained with one intensity (up to 100 instances per quadrant).

The Adult runs and the SHAP and Anchors runs of the other two datasets belong to a benchmark cohort described in [11]. That work compares explainers by quality measures; no result of it is repeated here. A companion manuscript under review studies explanation quality on correct and incorrect predictions with the same cohort; it does not study agreement between explainers.

The earlier LIME runs on German Credit and Breast Cancer explained a different sample of instances and stored only averages. *A new LIME cohort was therefore produced for this study:* for each dataset, model and seed, LIME explained exactly the instances stored in the corresponding SHAP run. The model binaries were regenerated from the training recipe in the repository, and the run was allowed only after the regenerated model reproduced the stored training metrics, the feature order and the stored label and prediction of every target instance.

3.2 Explainers

SHAP. TreeSHAP [16] was used for RF and XGB, in its interventional form and on the probability scale, and KernelSHAP [17] for LR, SVM and MLP, both with 50 background instances drawn from the training data. The stored value of a feature is its signed contribution to the predicted probability of

class 1, relative to the expected probability on the background data.

LIME. The tabular explainer [20] was used with 1000 perturbed samples, kernel width 3, ten selected features and *without discretisation of continuous features*. With this setting, LIME standardises each feature with the mean and the standard deviation of the training data, and the stored value of a feature is the coefficient of the local weighted linear model for the probability of class 1 on that standardised scale, that is, a local slope.

Anchors and DiCE (Adult, secondary analysis). Anchors [21] returns a rule; its feature set is the set of features in the rule. DiCE [18] returns counterfactual instances; its feature set is the set of features whose value changes. Neither is a signed attribution, so only feature sets are compared.

For each instance, a run stores the ten features with the largest absolute value, in that order. All measures below use these stored lists.

3.3 Agreement Measures

Let S_k and L_k be the sets of the k features with the largest absolute SHAP and LIME values for one instance. The primary measure is the top-5 Jaccard overlap [13]:

$$J_5 = \frac{|S_5 \cap L_5|}{|S_5 \cup L_5|}. \quad (1)$$

Three shared features out of five give $J_5 = 3/7$, and four give $J_5 = 4/6$. The top-10 overlap J_{10} is a sensitivity measure.

Rank concordance is Kendall's τ_b [14] computed on the union of the two top-10 lists; a feature absent from a list receives rank 11, and ties use midranks.

Sign agreement is the fraction of the features in $S_5 \cap L_5$, with both values different from zero, whose SHAP and LIME values have the same sign. For the post hoc analysis of Section 4.3, the LIME slope w_j of feature j was converted into a contribution relative to the training mean m_j :

$$c_j = w_j \frac{x_j - m_j}{s_j}, \quad (2)$$

where x_j is the value of the feature for the instance and s_j the standard deviation used by LIME. This

is the term of feature j in the local linear model of LIME, so c_j and the SHAP value are both contributions relative to a reference. The sign agreement was then recomputed with the sign of c_j .

As a reference, not as a threshold, the expected value of J_k for two independent random sets of k features out of p was computed exactly from the hypergeometric distribution. For $k = 5$ it is 0.026 on Adult, 0.047 on German Credit and 0.099 on Breast Cancer.

3.4 Pairing and Quality Control

Explanations were paired by dataset, model, seed, intensity and instance identifier. Only instances with a valid explanation from both methods and the same stored label and prediction entered the analysis. The source runs contained 278 records with an error and no explanation, mostly KernelSHAP on the SVM; these were excluded, and one SVM block had no usable SHAP run. Of the 87 blocks, 76 had identical sets of valid instances for the two methods; in the others the intersection was used, which left 4,936 valid records unpaired. No paired record had a different label or prediction in the two runs. The result is 31,411 paired explanations in 86 runs: 29,667 on Adult (74 runs), 1,060 on German Credit and 684 on Breast Cancer (six runs each). Kendall's τ_b was undefined for 0 pairs and sign agreement for 516 pairs without a shared non-zero feature.

3.5 Statistical Summary

The same test instances recur across models, seeds and intensities, so instances are not independent replicates. Each measure was first averaged within a run (dataset, model, seed, intensity). The point estimate of a group is the mean of its run means. The 95% interval is a percentile bootstrap with 2000 resamples in which random seeds, not instances, are resampled with replacement, and all runs of a sampled seed are kept together [5, 8]. There are five seeds on Adult and three on the other datasets, so the intervals describe variation between seeds and are not precise.

For RQ3, the difference between the mean J_5 of misclassified and of correctly classified instances was computed within each run that had at least ten instances in each group, averaged over intensities for each model and seed, and summarised by model. For RQ4, the Spearman correlation between disagreement ($1 - J_5$) and each of four quality values of the instance (fidelity and stability of the SHAP and of the LIME explanation) was computed within each run and averaged by seed. Fidelity [3] is the Pearson correlation, across features, between the absolute value of the attribution and the absolute change in the predicted probability when that feature alone is replaced by its training mean. Stability [1] is the mean pairwise cosine similarity between the explanations of 15 copies of the instance with small Gaussian noise added. Both were computed and stored with each explanation when the runs were made.

No hypothesis test was planned and no p -value is reported. The estimates for RQ2 to RQ4 are exploratory.

3.6 Registration and Deviations

The analysis plan, with the measures, the unit of analysis, the exclusion rules and the bootstrap settings, was committed to the repository before any agreement statistic was computed and before the new LIME cohort was run. Two refinements made before the analysis are recorded in the plan. One analysis is *post hoc*: after the planned sign agreement was inspected, it was broken down by predicted class, the constancy of the sign of each feature was measured, and the sign agreement was recomputed after converting the LIME slopes into contributions (Section 4.3). It is reported as such. The feature values needed for the conversion were reloaded with the code that produced the runs; the stored label of every paired instance matched the reloaded data.

4 Results

4.1 Agreement on the Top Features (RQ1)

Table 1 and Fig. 1 give the agreement estimates. On Adult, the mean top-5 overlap was 0.393 (95%

interval 0.377 to 0.410); on German Credit it was 0.387 (0.343 to 0.426). Both are below the value for three shared features out of five. On Breast Cancer the overlap was 0.714 (0.671 to 0.792), above the value for four shared features. In all three datasets the overlap was several times the expected overlap of random sets.

The order of the features agreed less than their identity. Kendall's τ_b was 0.126 on Adult and 0.122 on German Credit, and 0.517 on Breast Cancer. Extending the lists to ten features raised the overlap on Adult and German Credit (0.507 and 0.480) and lowered it slightly on Breast Cancer (0.669).

4.2 Model Family, Sampling Intensity and Dataset (RQ2)

On Adult, agreement differed between model families. The overlap was highest for logistic regression (0.485) and lowest for the multilayer perceptron (0.335); the intervals of these two families do not overlap. Rank concordance was close to zero for the multilayer perceptron (0.043) and for gradient boosting (0.028, with an interval that includes zero). On German Credit the overlap was higher for the random forest (0.464) than for gradient boosting (0.309); on Breast Cancer the two families were close.

The sampling intensity had no visible effect: pooled over models, the top-5 overlap on Adult was 0.395, 0.394 and 0.390 for 50, 100 and 200 instances per quadrant. Explaining more instances makes the estimate of agreement more precise; it does not change the agreement.

The dataset mattered most. Adult and German Credit, with many one-hot encoded features, gave similar and low agreement; Breast Cancer, with 30 numeric features, gave much higher agreement for the same two model families. Three datasets do not allow this difference to be attributed to a specific property of the data.

4.3 Agreement of Signs (Post Hoc)

The planned sign agreement on shared top-5 features was 0.717 on Adult, 0.511 on German Credit and 0.374 on Breast Cancer. The last value is below one half, on the dataset where the two methods agree most on which features matter. This led to the post hoc analysis in Table 2 and Fig. 2.

Sign agreement depended strongly on the predicted class. On Breast Cancer it was 0.977 for instances predicted as class 0 and 0.036 for instances predicted as class 1. On Adult it was 0.875 for class 1 and 0.506 for class 0, and the same pattern appeared in four of the five model families; the multilayer perceptron was the exception.

The reason is that the two methods return different quantities (Section 3.2). The LIME coefficient of a continuous feature is a local slope. The slope of a feature with a monotone effect has the same sign for most instances. A SHAP value is a contribution relative to the expected output: for the same feature it is positive for instances on one side of the reference and negative on the other. The two signs therefore coincide for the instances of one predicted class and differ for the other. The constancy measure confirms it: on Breast Cancer, a feature kept its majority sign in 0.993 of its LIME explanations and in 0.621 of its SHAP explanations (0.912 and 0.679 on German Credit; 0.850 and 0.757 on Adult, where most features are one-hot indicators).

The direct test confirms this explanation. When each LIME slope was converted into a contribution with Eq. (2), the sign agreement rose to 0.945 on Adult (95% interval 0.941 to 0.949), 0.960 on German Credit (0.953 to 0.964) and 0.990 on Breast Cancer (0.983 to 0.997), and the dependence on the predicted class disappeared: on Breast Cancer it was 0.991 for class 0 and 0.989 for class 1, and on Adult 0.927 and 0.958. It exceeded nine tenths for every model family (Table 2).

Sign agreement between SHAP and LIME without discretisation is thus not a measure of disagreement about the direction of an effect. It measures mostly on which side of the reference the instance lies. Once both explanations are

Table 1. Agreement between SHAP and LIME: mean of run means and seed-clustered 95% interval. Adult rows pool the three sampling intensities

Dataset	Model	Runs	Pairs	Top-5 Jaccard	Top-10 Jaccard	Kendall τ_b
Adult	LR	15	7,000	0.485 [0.464, 0.505]	0.617 [0.584, 0.640]	0.263 [0.221, 0.304]
	SVM	14	2,064	0.395 [0.388, 0.404]	0.516 [0.489, 0.539]	0.171 [0.148, 0.187]
	MLP	15	7,000	0.335 [0.326, 0.341]	0.441 [0.435, 0.448]	0.043 [0.029, 0.054]
	RF	15	6,603	0.389 [0.360, 0.415]	0.560 [0.530, 0.590]	0.129 [0.090, 0.168]
	XGB	15	7,000	0.363 [0.328, 0.399]	0.404 [0.384, 0.422]	0.028 [−0.020, 0.071]
	All	74	29,667	0.393 [0.377, 0.410]	0.507 [0.487, 0.525]	0.126 [0.100, 0.153]
German Credit	RF	3	523	0.464 [0.450, 0.471]	0.542 [0.519, 0.572]	0.237 [0.212, 0.251]
	XGB	3	537	0.309 [0.235, 0.381]	0.419 [0.395, 0.440]	0.008 [−0.101, 0.096]
	All	6	1,060	0.387 [0.343, 0.426]	0.480 [0.457, 0.506]	0.122 [0.055, 0.172]
Breast Cancer	RF	3	342	0.738 [0.678, 0.790]	0.665 [0.643, 0.691]	0.526 [0.494, 0.571]
	XGB	3	342	0.690 [0.612, 0.795]	0.673 [0.666, 0.681]	0.507 [0.457, 0.551]
	All	6	684	0.714 [0.671, 0.792]	0.669 [0.654, 0.682]	0.517 [0.475, 0.561]

Table 2. Sign agreement on shared top-5 features, overall and by predicted class (Cl.), with the LIME value as stored (slope) and after conversion into a contribution. The column All of the slope block is the prespecified estimate; the rest is post hoc

Dataset	Model	Slope			Contribution		
		All	Cl. 0	Cl. 1	All	Cl. 0	Cl. 1
Adult	LR	0.696	0.527	0.864	0.947	0.947	0.948
	SVM	0.829	0.332	0.956	0.952	0.918	0.960
	MLP	0.689	0.706	0.672	0.951	0.940	0.962
	RF	0.696	0.418	0.975	0.948	0.913	0.984
	XGB	0.682	0.443	0.916	0.925	0.911	0.939
	All	0.717	0.506	0.875	0.945	0.927	0.958
German Credit	RF	0.470	0.608	0.430	0.985	0.990	0.983
	XGB	0.553	0.716	0.494	0.934	0.933	0.934
	All	0.511	0.662	0.462	0.960	0.962	0.959
Breast Cancer	RF	0.377	0.975	0.041	0.997	1.000	0.995
	XGB	0.371	0.979	0.032	0.983	0.983	0.983
	All	0.374	0.977	0.036	0.990	0.991	0.989

contributions, the two methods agree on the direction of almost all the features that both place among the five most important.

4.4 Correct and Misclassified Instances (RQ3)

Table 3 and Fig. 3 give the difference in top-5 overlap between misclassified and correctly classified

instances. The direction was not consistent. For the multilayer perceptron, agreement was lower on misclassified instances (difference -0.059 , interval -0.069 to -0.049 , same direction in all five seeds). For the random forest it was higher (0.045 , 0.029 to 0.064 , all five seeds), and slightly higher for gradient boosting (0.014). For logistic regression and the support vector machine the difference was near zero. On German Credit the differences were small and negative, with three seeds. On Breast Cancer no run had ten misclassified instances, so the contrast could not be estimated. The largest difference in absolute value was that of the multilayer perceptron, small compared with the differences between model families.

4.5 Disagreement and Explanation Quality (RQ4)

Table 4 and Fig. 4 give the within-run Spearman correlation between disagreement and the four quality values. The clearest pattern concerns the fidelity of LIME: on Adult the correlation was negative for all five model families (from -0.45 to -0.19), and it was negative on German Credit (-0.26 and -0.24). The two methods disagreed more on the instances where the sizes of the LIME attributions followed less closely the effect of masking each feature. The correlations with the

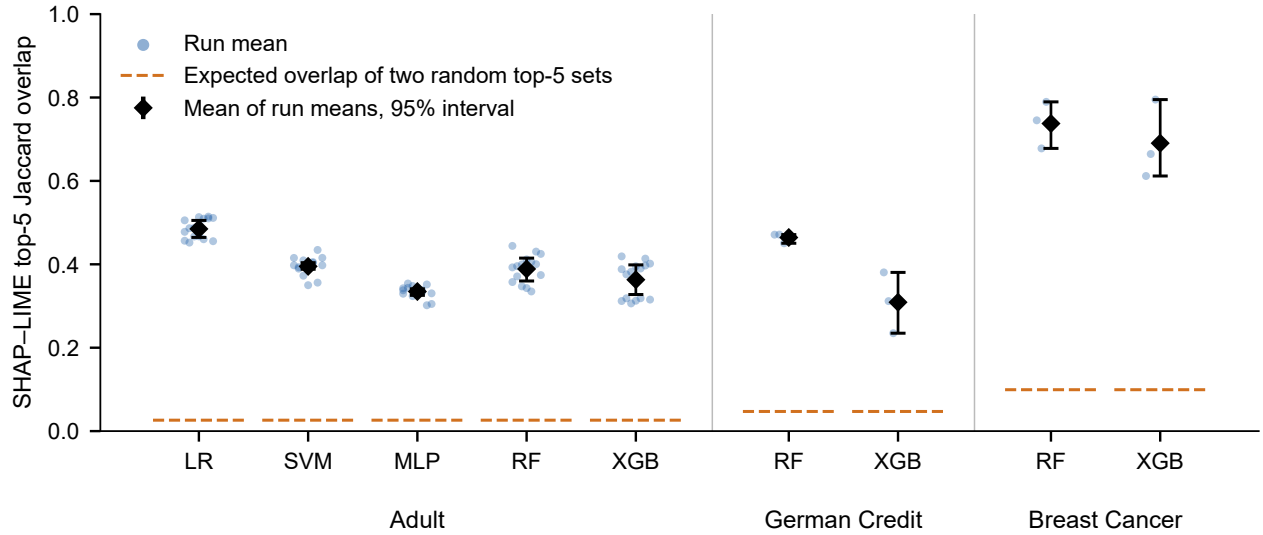


Fig. 1. Top-5 overlap between SHAP and LIME by dataset and model family. Each point is the mean of one run

Table 3. Top-5 overlap for correctly classified (C) and misclassified (M) instances. Diff. is M minus C; Neg. is the number of seeds with a negative difference; GC is German Credit

Data	Model	C	M	Diff.	95% interval	Neg.
Adult	LR	0.486	0.483	-0.003	[-0.023, 0.012]	2/5
Adult	SVM	0.388	0.387	-0.001	[-0.017, 0.020]	3/4
Adult	MLP	0.364	0.305	-0.059	[-0.069, -0.049]	5/5
Adult	RF	0.367	0.412	0.045	[0.029, 0.064]	0/5
Adult	XGB	0.356	0.370	0.014	[0.003, 0.022]	1/5
GC	RF	0.469	0.452	-0.017	[-0.036, 0.001]	2/3
GC	XGB	0.311	0.305	-0.005	[-0.010, -0.001]	3/3

fidelity of SHAP were closer to zero and of mixed sign.

For stability there was no common pattern. On Adult the correlations were small and mostly positive, except for the multilayer perceptron. On Breast Cancer with the random forest, disagreement was higher for instances with less stable explanations of either method (-0.56 and -0.54), with three runs. These are associations within runs; they do not show that low fidelity causes disagreement. Fidelity is computed from the absolute values of the attributions, so the difference in sign between

Table 4. Mean within-run Spearman correlation between disagreement ($1 - J_5$) and explanation quality. Fid.: fidelity; Stab.: stability

Dataset	Model	SHAP		LIME	
		Fid.	Stab.	Fid.	Stab.
Adult	LR	-0.01	0.09	-0.40	0.12
	SVM	-0.05	0.06	-0.45	0.12
	MLP	-0.06	-0.24	-0.19	-0.07
	RF	-0.22	0.23	-0.27	0.10
	XGB	-0.15	0.14	-0.40	0.11
German Credit	RF	-0.05	-0.06	-0.26	-0.10
	XGB	-0.04	0.02	-0.24	-0.15
Breast Cancer	RF	-0.06	-0.56	-0.28	-0.54
	XGB	0.22	-0.34	0.01	-0.27

slopes and contributions does not affect it, but their difference in size does.

4.6 Feature Sets of Anchors and DiCE (Secondary)

Table 5 gives the overlap between feature sets on Adult. The features in an Anchors rule overlapped more with the SHAP top-5 set (0.348) than with the LIME top-5 set (0.240). The features changed by

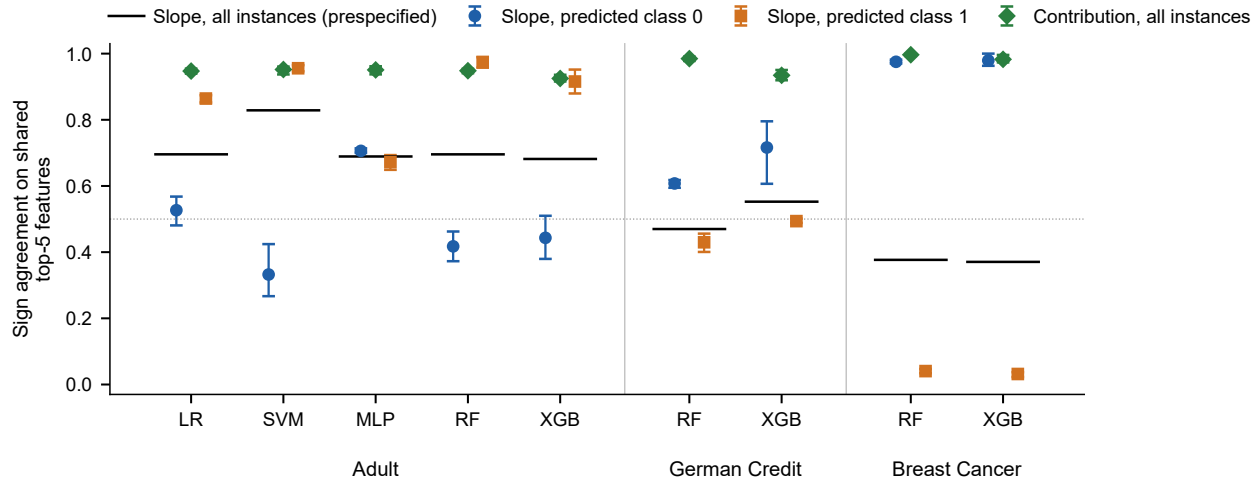


Fig. 2. Sign agreement between SHAP and LIME. Circles and squares: LIME value as stored (slope), by predicted class, with seed-clustered 95% intervals. Diamonds: all instances after the slopes were converted into contributions. The dotted line marks one half

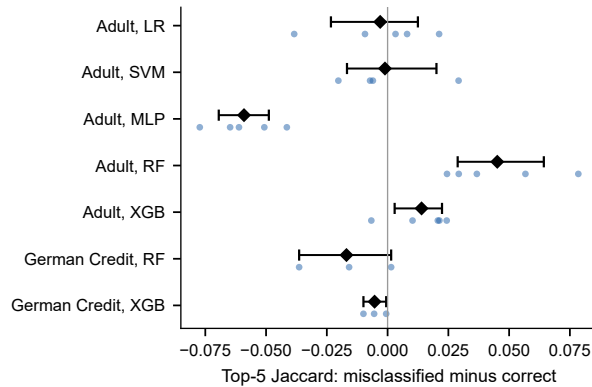


Fig. 3. Difference in top-5 overlap, misclassified minus correct. Points are seeds; diamonds are means with seed-clustered 95% intervals

Table 5. Jaccard overlap between feature sets on Adult (SHAP, LIME: top-5 features; Anchors: rule features; DiCE: changed features): mean of run means and seed-clustered 95% interval

Feature sets	Runs	Pairs	Mean	95% int.
SHAP vs. Anchors	56	14,584	0.348	[0.340, 0.356]
LIME vs. Anchors	57	19,095	0.240	[0.234, 0.247]
SHAP vs. DiCE	67	25,248	0.078	[0.068, 0.087]
LIME vs. DiCE	68	29,830	0.078	[0.074, 0.085]
Anchors vs. DiCE	53	18,578	0.068	[0.062, 0.075]

5 Discussion

The choice of explainer changes the explanation. On two of the three datasets, SHAP and LIME shared on average fewer than three of their five most important features, and the order of the features was almost unrelated for two model families. A user who reads a top-5 list would see a different list with the other method. This agrees with the disagreement reported by Krishna et al. [15] and extends it with the run as the unit of analysis: the result is stable across five seeds and three sampling intensities, so it is not an artefact of a particular sample of instances.

DiCE overlapped little with any other set (at most 0.078). These methods answer different questions, a sufficient condition and a minimal change, so a low overlap is not an error of either method; it shows that their features cannot be substituted for those of an attribution method.

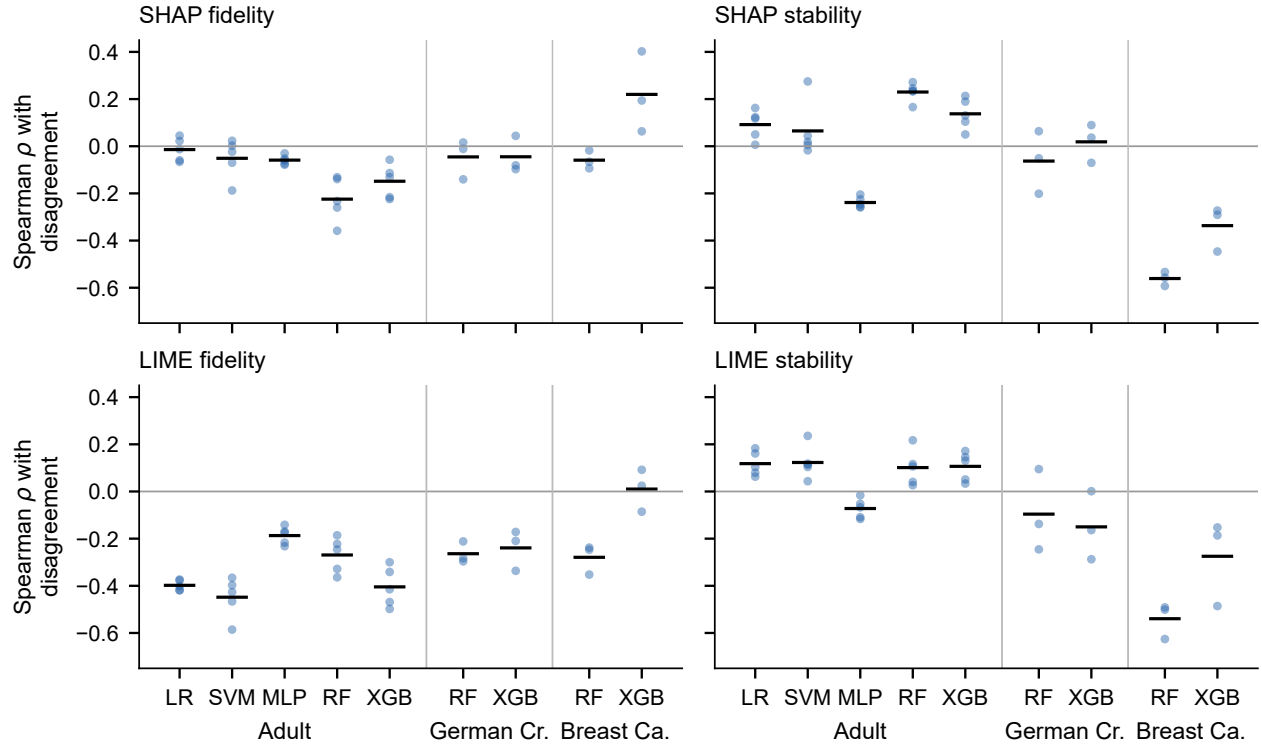


Fig. 4. Within-run Spearman correlation between disagreement and four quality values. Points are seeds; bars are means over seeds

Agreement is a property of the model and the data. The differences between model families and between datasets were larger than every other effect measured here. This is consistent with the view that the methods approximate the model in different neighbourhoods [10]: where the model is close to linear around the instance, as for logistic regression, the approximations coincide more. The high agreement on Breast Cancer indicates that low agreement is not inevitable. A report of explainer agreement should therefore state the model family and the dataset, and should not be generalised from one of them.

Signs need the same quantity. The sign analysis shows a risk for any study that compares the direction of attributions between methods. A LIME coefficient without discretisation is a slope; a SHAP value is a contribution relative to a reference [9, 17]. Their signs differed for almost every instance of one class on Breast Cancer, while the

two methods described the same relation between the feature and the output: after the slope was multiplied by the deviation of the feature from the training mean, the signs agreed for more than nine shared features out of ten on every dataset and model family. Before signs are compared, the two explanations should be brought to the same quantity in this way, or LIME should be used with discretisation, whose weights refer to the value of the instance. The disagreement problem, in these data, concerns which features are ranked first and in what order, not the direction of their effect. The measures of overlap and rank used for RQ1 to RQ4 depend on absolute values and are not affected by this difference of sign, although a slope and a contribution also differ in magnitude, which is one source of the low rank concordance.

Disagreement and correctness. Agreement was not consistently lower when the model was wrong. The direction depended on the model family

and the differences were small. Disagreement between explainers is therefore not a usable signal of a wrong prediction in these data.

Disagreement and LIME fidelity. Disagreement was higher where the fidelity of the LIME explanation was lower, that is, where the sizes of its attributions followed the single-feature masking effects less closely. Part of the disagreement may thus come from instances that LIME explains less well, not from two valid but different views of the model. The fidelity of LIME can indicate the instances for which the two methods are more likely to differ.

Limitations. The study covers three tabular binary datasets and one configuration of each explainer; other kernel widths, sample sizes or discretisation settings of LIME may change the estimates. There are five seeds on Adult and three on the other datasets, so the intervals are imprecise and the analyses of RQ2 to RQ4 are exploratory. Only the ten largest features of each explanation were stored, which limits the rank measure to that list. One-hot columns were treated as separate features; an analysis by original variable would give higher overlap. Fidelity and stability are proxy measures computed without ground-truth explanations. On Adult, each model was trained once, and for four of the five seeds part of the explained instances belong to its training data; the overall estimate for seed 42, whose evaluation partition is disjoint from the training data, lies within the range of the other four seeds. The conversion of slopes into contributions uses the training mean as reference, while SHAP uses the expected output on a background sample, so the two references are close but not identical. KernelSHAP failed for part of the SVM instances, so that family has fewer pairs. The models of the two smaller datasets were regenerated from the training recipe and validated against stored metrics and predictions, not compared byte by byte with the original binaries. For the Adult random forest, the stored binary does not reproduce all the predictions recorded in the runs; the analysis uses only the recorded outputs, and the paired SHAP and LIME records agree on every label and prediction. The sign analysis, including the conversion into contributions, is post hoc.

6 Conclusions and Future Work

For the same instance and model, SHAP and LIME agreed only partly on the most important features: the mean top-5 overlap was 0.393 on Adult, 0.387 on German Credit and 0.714 on Breast Cancer. Agreement depended on the model family and the dataset, not on the number of explained instances, and it was not consistently lower for misclassified instances. Disagreement was associated with a lower fidelity of the LIME explanation. The sign agreement between SHAP and LIME without discretisation followed the predicted class, because one method returns slopes and the other contributions. Once the slopes were converted into contributions, the two methods agreed on the sign of 0.945 to 0.990 of the shared top features. Sign agreement should not be used as a measure of disagreement unless both explanations are first expressed as the same quantity.

Two practical consequences follow. An explanation shown to a user should name the explainer, since another explainer would often show other features. And a comparison of explainers should check, before any measure is computed, what each stored value means.

Future work will compare the ranks of the two methods on a common quantity, repeat the analysis by original variable, vary the LIME configuration, and add datasets to separate the effect of the number and type of features from that of the model family.

Data Availability

The three datasets are public [2, 12, 25]. The code, the per-instance run outputs, the new LIME cohort, the registered analysis plan and every result file are available at <https://github.com/jonnabio/xai-eval-framework> (release paper-e-cys-2026-10-03), archived at <https://doi.org/10.5281/zenodo.23130949>. Every number, table and figure in this article is generated from those files by the scripts described in docs/reports/paper_e/README.md.

References

1. **Alvarez-Melis, D., Jaakkola, T. S. (2018).** On the robustness of interpretability methods. ICML Workshop on Human Interpretability in Machine Learning (WHI 2018). <https://arxiv.org/abs/1806.08049>.
2. **Becker, B., Kohavi, R. (1996).** Adult [dataset]. UCI Machine Learning Repository. DOI: [10.24432/C5XW20](https://doi.org/10.24432/C5XW20).
3. **Bhatt, U., Weller, A., Moura, J. M. F. (2020).** Evaluating and aggregating feature-based model explanations. Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI), pp. 3016–3022. DOI: [10.24963/ijcai.2020/417](https://doi.org/10.24963/ijcai.2020/417).
4. **Breiman, L. (2001).** Random forests. Machine Learning, Vol. 45, No. 1, pp. 5–32. DOI: [10.1023/A:1010933404324](https://doi.org/10.1023/A:1010933404324).
5. **Cameron, A. C., Miller, D. L. (2015).** A practitioner's guide to cluster-robust inference. Journal of Human Resources, Vol. 50, No. 2, pp. 317–372. DOI: [10.3368/jhr.50.2.317](https://doi.org/10.3368/jhr.50.2.317).
6. **Chen, T., Guestrin, C. (2016).** XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794. DOI: [10.1145/2939672.2939785](https://doi.org/10.1145/2939672.2939785).
7. **Doshi-Velez, F., Kim, B. (2017).** Towards a rigorous science of interpretable machine learning. arXiv preprint arXiv:1702.08608. <https://arxiv.org/abs/1702.08608>.
8. **Efron, B., Tibshirani, R. J. (1993).** An Introduction to the Bootstrap. Chapman & Hall, New York.
9. **Garreau, D., von Luxburg, U. (2020).** Explaining the explainer: A first theoretical analysis of LIME. Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS), Vol. 108, pp. 1287–1296. <https://proceedings.mlr.press/v108/garreau20a.html>.
10. **Han, T., Srinivas, S., Lakkaraju, H. (2022).** Which explanation should I choose? A function approximation perspective to characterizing post hoc explanations. Advances in Neural Information Processing Systems, Vol. 35, pp. 5256–5268. DOI: [10.52202/068431-0380](https://doi.org/10.52202/068431-0380).
11. **Herrera-Vásquez, J., Herrero-Uceda, M. (2026).** A framework for rigorous evaluation of model-agnostic explainability methods: multi-metric statistical benchmarking, operational protocol, and reproducibility. Revista de Investigación Multidisciplinaria Iberoamericana (RIMI), No. 3. DOI: [10.69850/rimi.vi3.307](https://doi.org/10.69850/rimi.vi3.307).
12. **Hofmann, H. (1994).** Statlog (German credit data) [dataset]. UCI Machine Learning Repository. DOI: [10.24432/C5NC77](https://doi.org/10.24432/C5NC77).
13. **Jaccard, P. (1912).** The distribution of the flora in the alpine zone. New Phytologist, Vol. 11, No. 2, pp. 37–50. DOI: [10.1111/j.1469-8137.1912.tb05611.x](https://doi.org/10.1111/j.1469-8137.1912.tb05611.x).
14. **Kendall, M. G. (1945).** The treatment of ties in ranking problems. Biometrika, Vol. 33, No. 3, pp. 239–251. DOI: [10.1093/biomet/33.3.239](https://doi.org/10.1093/biomet/33.3.239).
15. **Krishna, S., Han, T., Gu, A., Wu, S., Jabbari, S., Lakkaraju, H. (2024).** The disagreement problem in explainable machine learning: A practitioner's perspective. Transactions on Machine Learning Research. <https://arxiv.org/abs/2202.01602>.
16. **Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., Lee, S.-I. (2020).** From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence, Vol. 2, No. 1, pp. 56–67. DOI: [10.1038/s42256-019-0138-9](https://doi.org/10.1038/s42256-019-0138-9).
17. **Lundberg, S. M., Lee, S.-I. (2017).** A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, Vol. 30, pp. 4765–4774.
18. **Mothilal, R. K., Sharma, A., Tan, C. (2020).** Explaining machine learning classifiers through diverse counterfactual

explanations. Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, pp. 607–617. DOI: [10.1145/3351095.3372850](https://doi.org/10.1145/3351095.3372850).

19. **Neely, M., Schouten, S. F., Bleeker, M. J. R., Lucic, A. (2021).** Order in the court: Explainable AI methods prone to disagreement. ICMML Workshop on Theoretic Foundation, Criticism, and Application Trend of Explainable AI. <https://arxiv.org/abs/2105.03287>.
20. **Ribeiro, M. T., Singh, S., Guestrin, C. (2016).** Why should I trust you?: Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1135–1144. DOI: [10.1145/2939672.2939778](https://doi.org/10.1145/2939672.2939778).
21. **Ribeiro, M. T., Singh, S., Guestrin, C. (2018).** Anchors: High-precision model-agnostic explanations. Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, pp. 1527–1535. DOI: [10.1609/aaai.v32i1.11491](https://doi.org/10.1609/aaai.v32i1.11491).
22. **Roy, S., Laberge, G., Roy, B., Khomh, F., Nikanjam, A., Mondal, S. (2022).** Why don't XAI techniques agree? Characterizing the disagreements between post-hoc explanations of defect predictions. 2022 IEEE International Conference on Software Maintenance and Evolution (ICSME), pp. 444–448. DOI: [10.1109/ICSME55016.2022.00056](https://doi.org/10.1109/ICSME55016.2022.00056).
23. **Rudin, C. (2019).** Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence, Vol. 1, No. 5, pp. 206–215. DOI: [10.1038/s42256-019-0048-x](https://doi.org/10.1038/s42256-019-0048-x).
24. **Slack, D., Hilgard, S., Jia, E., Singh, S., Lakkaraju, H. (2020).** Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods. Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, pp. 180–186. DOI: [10.1145/3375627.3375830](https://doi.org/10.1145/3375627.3375830).
25. **Wolberg, W., Mangasarian, O., Street, N., Street, W. (1993).** Breast cancer Wisconsin (diagnostic) [dataset]. UCI Machine Learning Repository. DOI: [10.24432/C5DW2B](https://doi.org/10.24432/C5DW2B).

*Article received on DD/MM/YYYY; accepted on DD/MM/YYYY.
Corresponding author is Jonathan Herrera-Vásquez.